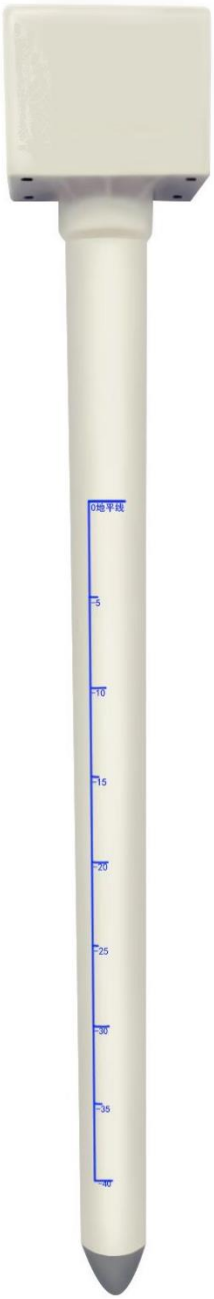
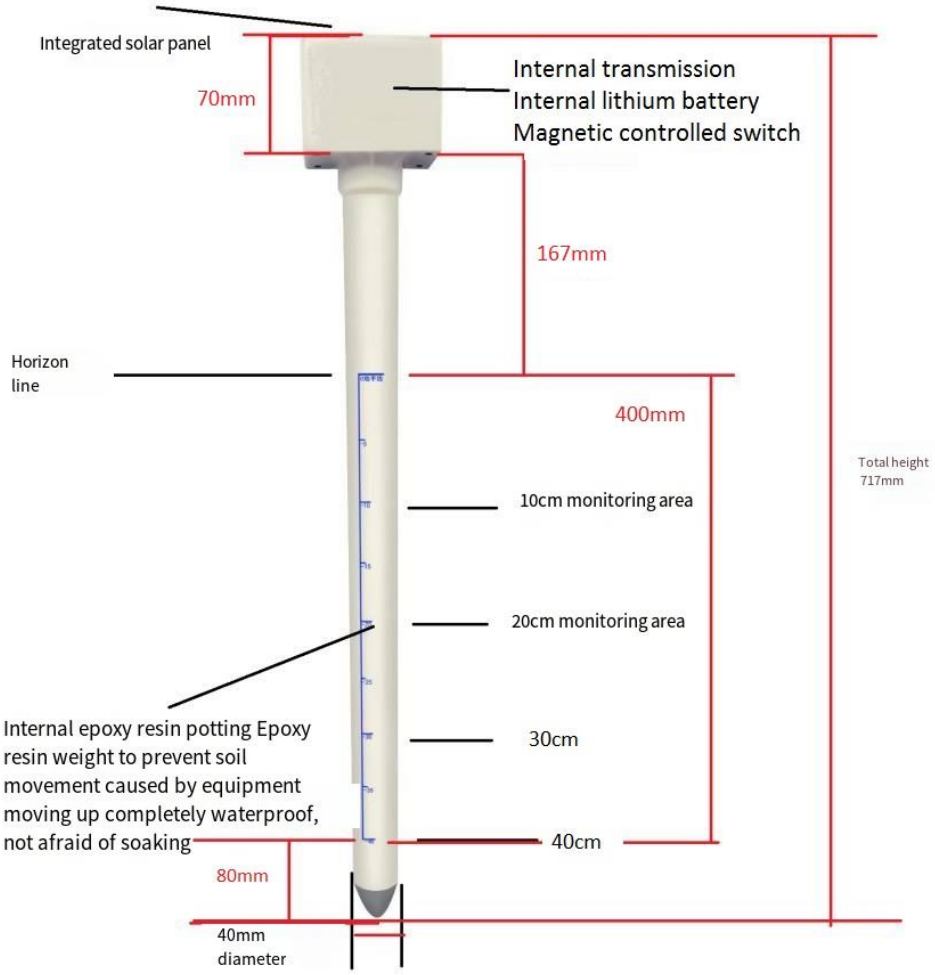
Tubular soil sensor

Product instruction manual





Four layer tube moisture dimension chart

1. Scope of application

Tubular soil sensor is a newly developed product of our company. This sensor adopts high-precision chip and advanced algorithm as well as IOT technology, which can measure multi-depth soil parameters, RS485 output or 4G network uploaded to the data center.

The product can be widely used in real-time monitoring of soil temperature and humidity in farmland, forest area, grassland and irrigation area. It can also be used in water conservancy and land disaster industries to provide data support for monitoring natural disasters such as landslides and debris flows.

2. Functional Features

- The product is designed to be flexible and can measure the temperature and humidity of the soil at any depth between 10-80cm (generally 10cm layer). Default for 4 layer, 5 layer, 8 layer standard tube. Support customization.
- It consists of sensing, collecting, transmitting and power supplying parts, and is an integrated design for easy installation.
- With domestic version, international version of the data transmission program, convenient for users outside the mainland. (Please inform the sales staff in advance of the international version)
- The collector configuration software adopts WeChat applet to connect with cell phone through Bluetooth, which can be used to manipulate the device in any Android or Apple cell phone end equipped with WeChat APP, and realize the deep interaction, configuration and remote sending between cell phone APP and the device. (Some functions are currently open to users)
- Sensor built-in high-efficiency lithium battery, RTU with low-power consumption design, fully charged state in the case of continuous rainy days can work for more than 180 days.
- Adopt magnetic switch design, can be used with the magnet supplied with the collector for switching control. Prevent the station in the field occurs human error shutdown situation.
- Adopting cloud platform storage technology, the data sent each time can be saved to the autonomous cloud platform or the third-party platform that meets the *Hydrological Monitoring Data Communication Statute SL651-2014.* It supports MQTT service docking to the third-party platform, and supports protocol customization. Default API interface docking.
- Other features: battery indicator, network location.
- Expandable elements: GPS positioning for more accurate positioning. Expandable parameters such as temperature, humidity, air pressure, dew point

temperature and body temperature.

- Expandable function: open the user's secondary calibration function (function of cubic term + quadratic term + primary term + constant), providing more accurate data for professional researchers. Users can provide data, our company to assist the coefficient calculation of the function.

3. Technical Parameters

Measured parameter: volumetric soil water content (soil moisture)

- Unit: % (m^3/m^3), or %Vol (volumetric water content)
- Humidity range: 0~100%Vol; Resolution:0.01%Vol
- Accuracy: ≤3% Vol within the valid range
- Measuring area: 90% of the influence is within the cylindrical measuring carrier with a diameter of 10 cm around the sensor
- Precision drift: none
- Probability of linear dispersion deviation of the sensor: 1%.

Measured parameters: soil temperature

- Soil temperature range: -40~+60℃; Resolution:0.01℃.
- Accuracy: ±0.5℃
- Stabilization time: approx 1 second after power-on
- Response time: Response goes to steady state within 1 second
- Sensor operating voltage: sensor input 5-24V DC
- Sensor operating current: quiescent current 4mA, acquisition current 35mA
- Sensor waterproof level: IP68
- Working Temperature: -40℃~+80℃
- RTU power consumption: standby 10μAh, transmission 100mAh.
- Transmission: default 4G output to own platform. Optional RS485 and other outputs, optional third-party docking.
- Actual power supply capacity of solar panel: ≤0.6W
- RS485 output power supply: DC5-24V, current less than 25mA.

4. Device Communication Protocols and Wiring

- *Hydrological Monitoring Data Communication Statute SL651-2014*; TCP protocol; other third-party customized protocols, MQTT, API interfaces, etc.
- The RTU and the sensing part are connected by internal cables. The

IoT card costs 3 years (one year for the international version).

- Solar panel Internal wiring of lithium batteries for power.

5. Selection of collection sites

The selection of representative plots for moisture monitoring should take into account their geomorphology, soil, meteorological and hydrogeological conditions, as well as the representativeness of species and plants. The scope of collection can be based on the county level, and a few representative areas can be selected at the county level (the reference basis for representativeness can be based on the categories of landforms and the extent of the areas of priority crops).

If you are located in a hilly area, you should set the testing point in a plot with a small slope ratio drop and a large area, and you should not collect in the bottom of a ditch or in a plot with a large slope.

Representative plots in plain areas should be collected on flat and non waterlogging plots

Detailed collection plots should be more than 10 meters away from the edge of representative plots, roadsides and in level plots, avoiding low-lying places prone to waterlogging plots, and maintaining a distance of more than 20 meters from ditches and flood channels to avoid the impact of seepage of water from ditches on soil water content.

For the collection of plots within the hydrological station, it is strictly forbidden to be in the middle of the vegetable field within the station to avoid the impact of watering on the collection accuracy. It is recommended to choose the collection point in a more open location, not close to houses or fences;

6. Sensor installation and data reporting

6.1 Installation Precautions

（1） Observe the data through the platform and confirm that the data is normal.

（2） Preparation of tools for installation (basin, water, shovel (or earth auger), sieve with 2-4 mm aperture)

（3） Holes shall be punched vertically downward;

（4） Drill holes to a suitable depth and note if there are more stones. There are many stones and it is recommended to relocate;

（5） Backfill the mud and insert the sensor vertically. (very important and fundamental to the accuracy of the response data);

（6） Verify that the sensor is mounted at a depth where the horizon is level with the ground.

6.2 Sensor Installation

（1） Select a more level installation location and make a hole in the ground directly with an earth auger:

（2） Remove the blue tube cap from the bottom of the tube test soil auger to take out the handles, and screw the two handles into the ends of the pair of wires in a clockwise direction; as shown below:

（3） Insert the pipe soil auger vertically along the center hole of the flange, press down with appropriate force. At the same time, rotate both hands left and right parallel to each other until it reaches the required depth (this process can feel whether there are stones in the soil, and if there are too many stones, you need to re-select the site), and in the process of punching, every maximum depth of 10cm, you need to raise the soil auger to fling the soil out of the pipe with force, and you can also use the extended screwdriver to take the hammer (can not use the hammer to hit the pipe body, so as not to cause deformation of the pipe body) to knock the lower part of the soil auger pipe body to vibrate the soil out or from the bottom of the screwdriver to dig the soil until all the soil

falls, with a ruler to measure the depth of the hole should be deeper than the sensor's measurement of the depth of 7-8cm; the following figure:

（4） The soil removed from the perforations is sieved and made into a slurry; as follows:

（5） Fill the drilled hole with slurry to a depth of at least 2/3 of the hole, as shown below.

（6） Simply insert the sensor into the backfilled mud ground hole with the sensor labeled with the horizon line flush with the ground. Figure below:

6.3 Voice recognition of the collector

“Beep” for power on;

Two “beeps” means it is off;

7. System setup, data reporting, data queries

7.1 Equipment switching and configuration

（1） Scan the QR code with your cell phone to see the data, and use the magnetic peg to power up the sensor (the default state of power on in China, restart after brushing the magnetic, power on the data).

（2） If you need to configure, you can enter the small program after clicking

on the “click to connect Bluetooth” 2 seconds after the interface will appear in the lower part of the button, click on the button will prompt the “connection is successful”, (this feature has the option to open, generally configured in accordance with the user’s requirements before leaving the factory)

7.2 Data reporting

Each time the sensor is turned on, the RTU automatically reports a test message to the receiving center. After the initial installation is completed, the sensor can be turned on for about 1 minute to view the last reported data.

7.3 Platform operation and data query

You can log in to the platform to view real-time data, historical records, data export and more.

8. Problem Determination and Solution

Description of the problem	Solutions
1. No start-up beep	● Go to a sunny spot to light the solar panel and charge the sensor for at least one day; ● After charging, it still can not be turned on.It can be returned to the factory for inspection and repair.
2. Unable to collect data	● Whether the sensor has been inserted into the soil or is in the measurement carrier; ● If the cause cannot be determined, it can be returned to the factory for inspection and repair.
3. Inability to report data (No communication)	● Check whether the cell phone has a signal, if it is in the state of no signal RTU can not report data; ● It can be returned to the factory for service after ruling out other causes; ● No communication for RS485 version. Check that the power supply and RS485 wiring, communication parameters are configured correctly and communication commands are sent correctly.

Note: This hardware and WeChat small program, PC may be updated at any time, if no

special circumstances without notice!

Selection List:

HCS Series Catheterized Soil Senosr				
Selection table				
Serial number	**Name**	**Model number**	**Output**	**Note**
1	Conduit soil monitor	HCS-4	RS485	5 Meter Cable
2	Conduit soil monitor	HCS-5	RS485	5 Meter Cable
3	Conduit soil monitor	HCS-8	RS485	5 Meter Cable
4	Conduit soil monitor	HCS-4/G	4G	3-year flow
5	Conduit soil monitor	HCS-5/G	4G	3-year flow
6	Conduit soil monitor	HCS-8/G	4G	3-year flow
Note: Standard 10cm layer, customized protocols and customized layers are available.				

Modbus RTU communication protocol

MODBUS RTU communication protocol (for Catheterized Soil Moisture Sensor. **Protocol supports customization.**)

Baud rate: 9600; Data bits: 8; Stop bits: 1; Parity bit: none

CRC Notes:

In all the following descriptions, the two bytes of CRC16 in the MODBUS RTU protocol, according to MODBUS regulations: the low byte comes first and the high byte comes second.

1. Return error code provisions:

The sensor does not return the error code for the reception of error commands (including CRC16 validation errors). The host computer can consider that the instruction has failed when no return data is received 100ms after the instruction is issued, and can resend the instruction.

2. Standard MODBUS Register Description

Special Note: The number or length of registers in MODBUS commands are in two-byte 16-bit units (high byte first, low byte second), not single-byte 8-bit units.

The user should ensure that the range of the two parameters, address and number of registers in the command is within the range specified by the system. If they are out of range, the sensor output results will be unpredictable. The user should ensure that the MODBUS commands comply with the requirements of this manual in the design of the host computer software.

3. Internal registers: read with function code 03; write with function code 06.

Order	Register address	Parameter description	Note
	Ox0100	Device address, default ox01.	fill out or in (information on a form)
1 layer	ox0020	Soil volumetric water content 1 layer (10 cm depth). A hexadecimal number magnified 100 times, e.g. 0x08D0 means 2256/100=22.56%Vol	read-only
	ox0021	Soil temperature 1 layer, (10cm depth), magnified 100 times hexadecimal number, e.g. 0x08D0 indicates 2256/100 = 22.56°C. Negative numbers are expressed as complements.	read-only
2 layer	ox0022	Soil moisture content by volume2	read-only
	ox0023	Soil temperature2	read-only
3 layer	ox0024	Soil moisture content by volume3	read-only
	ox0025	Soil temperature 3	read-only
4 layer	ox0026	Soil moisture content by volume4	read-only
	ox0027	Soil temperature 4	read-only
5 layer	ox0028	Soil moisture content by volume 5	read-only
	ox0029	Soil temperature 5	read-only
6 layer	ox002A	Soil moisture content by volume6	read-only
	ox002B	Soil temperature 6	read-only
7 layer	ox002C	Soil moisture content by volume 7	read-only
	ox002D	Soil temperature 7	read-only
8 layer	ox002E	Soil moisture content by volume 8	read-only
	ox002F	Soil temperature 8	read-only

Note: Address data is represented in hexadecimal. For example: address 20 (decimal) read/write value is 0x14.